

Daily Problem

November 30, 2025

Find:

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{1 + e^x} dx$$

Solution

Let $x = -t$. Then we can get:

$$\begin{aligned} I &= \int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} \frac{\cos(-t)}{1 + e^{-t}} (-dt) \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos t}{1 + e^{-t}} dt \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{1 + \frac{1}{e^x}} dx \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{e^x \cos x}{e^x + 1} dx \end{aligned}$$

$$\begin{aligned} 2I &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{1 + e^x} dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{e^x \cos x}{1 + e^x} dx \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x + e^x \cos x}{1 + e^x} dx \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x(1 + e^x)}{1 + e^x} dx \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x dx \end{aligned}$$

Now, we have:

$$2I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x dx$$

therefore

$$I = 1$$